

# Enhancing CLIP for Fine-Grained Classification and VQA with VLM-Guided Attention Steering

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## 1 Introduction

Contrastive Language-Image Pre-training (CLIP) excels in many multimodal tasks that require a broad understanding of semantic relationships between images and text. This is accomplished through a large scale contrastive pre-training objective, mapping matching image-text pairs close together in a shared embedding space, while pushing mismatched pairs apart. (Radford et al., 2021) However, CLIP displays less efficacy when a task requires the ability to discriminate between fine-grained visual details within an image, such as fine-grained image classification or visual-question answering. A potential reason for this performance reduction is CLIP’s visual encoder under emphasizing regions or attributes within the image that are critical for capturing lower level saliency, in favor of capturing global semantic information. Current state of the art VLMs have exhibited strong multimodal reasoning, visual-textual grounding, and semantic understanding across diverse domains, with an ability to provide expert level guidance on a wide variety of vision-language tasks. In order to enhance CLIP’s fine grained recognition and VQA ability, We seek to inject VLM guidance of discriminative, class based visual attributes into CLIP’s vision transformer backbone through an inference time attention steering approach, requiring no additional training time and minimal added complexity.

## 2 Guidance Pipeline For Classification

The first step in our methodology is an offline pipeline to generate attributes that identify discriminative visual features for each class within a task-specific classification dataset. We provide a vision-language model (Qwen2.5-VL-7B) with a representative image of each class and the class name, and ask the model to produce a structured list of three salient visual characteristics about that class. To

guide the VLM toward useful outputs, we provide two in-context training examples drawn from our prior experiments. These samples include an image, a list of general, non class-specific characteristics in which attention steering lowered overall classification accuracy in comparison to unsteered CLIP, and a list of manually selected, class-specific queries that increased accuracy. Each image in the dataset is then passed through a text-to-region grounding detector (Grounding-dino), in which bounding box coordinates for each region matching any of the three VLM-generated, class specific attributes are collected. We set the box-confidence threshold to 0.25, and transfer the bounding boxes into CLIP’s coordinate space by applying the same resize and center-crop transformations used by CLIP’s image preprocessing mechanism. We then convert these localized regions into patch-level steering targets. Since CLIP ViT-B-16 represents the image as a 14×14 grid of patch tokens, we select all patches whose area overlaps by at least 75 percent with one of the generated bounding boxes. We illustrate the results of this process in Figure 1, which shows selected patches overlaid in green on representative images from two classes. While grounding accuracy remains a limitation, as seen by the selection of several background patches in the two provided examples, we hope to provide a baseline for future work exploring stronger text-to-region grounding models and patch-selection strategies.

## 3 Methodology

Following the offline preprocessing stage, each test image is associated with a set of guidance patches on the 14×14 CLIP ViT-B-16 patch grid. We use these stored patch selections to bias attention through a two-step procedure. First, a one-time, dataset-specific head-profiling pass identifies the attention heads most receptive to the steering signal. Second, attention steering is applied to the



Figure 1: Examples of grounded patch regions selected for attention steering.

selected heads at inference time by downweighting the attention scores assigned to unselected patch tokens.

### 3.1 Head Profiling

Similar to prior work on post-hoc attention steering (Zhang et al., 2024), we find that steering all attention heads performs worse than a targeted subset. Therefore, we begin our methodology by steering each head between transformer blocks 4-12 individually on a disjoint subset of our test data. Selecting a range that targets mid to later layers ensures that we allow the model to build an initial, low-level visual representation of the image before steering. We select the top-K performing heads to steer simultaneously at inference time, and treat K as a hyperparameter determined through a validation grid sweep. To avoid biasing head selection toward any particular class, the profiling subset is constructed with a representative number of images sampled from each class.

### 3.2 Inference-Time Attention Steering

A ViT block computes self-attention as

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right) \in \mathbb{R}^{S \times S},$$

where, for CLIP’s ViT-B/16 backbone,  $S = 197$  is the sequence length, consisting of one [CLS] token and 196 spatial patch tokens.  $Q$ ,  $K$ , and  $V$  denote the query, key, and value matrices for a given attention head,  $d_h$  is the head dimension, and  $A_{q,k}$  is the attention weight from query token  $q$  to key token  $k$ . The output of the attention head is

$$H = AV \in \mathbb{R}^{S \times d_h},$$

returning a refined, context-aware representation of each token in the input sequence.

Given a set of selected indices  $\mathcal{G} \subseteq \{1, \dots, 196\}$  produced by our patch-selection pipeline, we bias

the model toward the selected spatial patches by reweighting the attention distribution at each selected head. Following the post-hoc steering formula of (Zhang et al., 2024), we partition the keys into a preserved set  $\mathcal{G} \cup \{0\}$ , where index 0 denotes the [CLS] token, and the remaining unselected patch tokens form the complement  $\bar{\mathcal{G}}$ . For a given query row  $q$ , we define the reweighted attention as

$$\tilde{A}_{q,k} = \begin{cases} \frac{A_{q,k}}{C_q}, & k \in \mathcal{G} \cup \{0\}, \\ \frac{\alpha A_{q,k}}{C_q}, & k \notin \mathcal{G} \cup \{0\}, \end{cases}$$

where

$$C_q = \sum_{k \in \mathcal{G} \cup \{0\}} A_{q,k} + \alpha \sum_{k \notin \mathcal{G} \cup \{0\}} A_{q,k}$$

is a per-row normalizer. The coefficient  $\alpha \in [0, 1]$  controls the steering strength, with smaller values of  $\alpha$  concentrating more attention mass on the selected keys.

We introduce two significant adjustments to adapt the work of Zhang et al. into the visual domain. First, we choose to add the CLS token to the preserved set rather than the complement. The CLS token serves as a global image representation that is updated across transformer blocks through self attention, and is ultimately used as the final image embedding to be transformed into CLIP’s shared vision-text embedding space. Therefore, given that the CLS token can not be selected by our processing pipeline, down-weighting attention to it at all selected heads would likely weaken CLIP’s ability to preserve a running summary of the image, and does not align with our overall objective.

Second, we restrict the steering operation to just the [CLS] query row, leaving the attention distributions of all patch tokens unchanged. This allows patch tokens to better preserve their local contextual information, which may be especially important when discriminative visual evidence depends not only on the object itself, but also on surrounding scene context. For example, two bird species may share similar visual characteristics, but typically appear in different environments.

## 4 Related Work

Recent work has sought to address the limitations of CLIP’s fine-grained recognition and VQA ability, as well as methods for steering transformer attention toward task-relevant tokens.

## 4.1 Fine Grained Vision-Language Alignment

FG-CLIP (Xie et al., 2025) improves CLIP’s fine-grained classification ability through the introduction of a large-scale fine-tuning dataset. The dataset includes 1.6 billion long, descriptive caption image pairs, designed to allow CLIP to recognize fine-grained image details. Additionally, over 40 million region level descriptions localized from 12 million images, with 10 million hard negative examples are designed to refine CLIP’s ability to distinguish between subtle visual differences within closely related classes. Results show that CLIP models fine-tuned on the proposed dataset outperform standard CLIP, as well as other previous state of the art methods in various image-language understanding benchmarks. Our methodology pursues a similar objective to FG-CLIP, improving CLIP’s sensitivity to fine-grained visual attributes, but does so through inference-time attention steering rather than large-scale fine-tuning.

AG-CLIP (Ahmad et al., 2026) introduces an attribute-guided framework for fine-grained image classification. The authors introduce an LLM-guided pipeline to mine relevant visual attributes from class descriptions, and then construct prompts for an open-vocabulary detector to localize relevant regions within a given image. These regional representations are fused together using a fine-tuned attribute-fusion module, and are later combined with the global image representation output from CLIP’s visual backbone through a lightweight transformer fusion block. Our work shares AG-CLIP’s motivation of utilizing a language model to generate class-specific attributes, as well as employing a text-to-region model to ground attributes within test images. However, unlike AG-CLIP, we increase the robustness of the generation pipeline by providing the generating VLM with several in context learning examples and a representative image of each class. Our approach also seeks a less computationally expensive method to inject VLM guidance into CLIP’s visual backbone.

## 4.2 Attention Steering for Transformers

Tell Your Model Where to Attend: Post Hoc Attention Steering for LLM’s (Zhang et al., 2024) introduced an inference time method for steering a language models attention toward user-selected words in an input prompt. Given a set of token indices the user wants the model to attend to, PASTA selects a subset of attention heads, and within each,

scales down the attention weights of unselected tokens by a hyperparameter  $\alpha < 1$ . Overall, the PASTA method results in increased task following ability for language models. Our experiment closely follows the methodologies introduced in PASTA, adapting them to the visual domain. We also follow a similar head profiling step before attention steering, and opt to down weight the attention scores of unselected tokens in order to increase numerical stability. However, our methodologies differ extensively in the token selection domain. Unlike PASTA, we seek VLM guidance during the patch selection phase, with hopes that an automated, expert-level patch selection approach will further increase the effectiveness of the method.

# 5 Results and Discussion

We focus our experiments on the CUB-200-2011 dataset, a fine-grained bird classification benchmark containing 200 distinct bird species. CUB-200-2011 is especially challenging because many bird species share highly similar global shapes and differ only through localized visual attributes such as wing patterns, head color, or beak shape. This makes the dataset a strong testing ground for attention steering through our VLM guidance pipeline, since improvements should primarily come from emphasizing the fine-grained regions that distinguish one species from another. CUB-200-2011 also includes many images that are not tightly cropped around the bird unlike similar classification datasets, leaving surrounding background and environmental context visible. This allows us to observe whether attention steering can emphasize discriminative bird regions while still preserving useful scene information, such as water or ground terrain that may provide secondary cues for classification.

## 5.1 Experiments

We reserve 500 images for our head profiling phase, and another 2500 images for testing. Similar to related work, we augment each class name with a prompt "A photo of a {class\_name}" to further align with CLIP’s pretraining objective.

| Method                     | Acc. (%) | $\Delta$ |
|----------------------------|----------|----------|
| Plain CLIP (CLIP ViT-B/16) | 47.68    | –        |
| Steered CLIP               | 48.00    | +0.32    |

Table 1: Accuracy on CUB-200-2011 (N=2500).  $\Delta$  is relative to Plain CLIP. Steered CLIP uses the top-10 profiled heads with  $\alpha = 0.3$ .

Our steering methodology results in only a modest improvement in classification accuracy relative to plain CLIP, with a 0.32 percentage-point increase. Additionally, steering leads to 90 changed predictions, corresponding to approximately 4 percent of the test set. To better understand these changes, we examine the classes most positively and negatively affected by steering relative to plain CLIP.



(a) Swainson Warbler



(b) Vesper Sparrow

Figure 2: Representative examples from the classes most positively and negatively affected by attention steering.

The Swainson’s Warbler, shown on the left of Figure 2, has steering improve classification accuracy by 30 percentage points across its 10 test samples relative to plain CLIP, while the Vesper Sparrow, shown on the right, decreases by 30 percentage points. Both classes are visually similar to each other and to other bird species in the dataset, suggesting that steering could have the strongest effect near class boundaries, where small shifts in visual emphasis can produce either corrected predictions or new misclassifications.

We also examine how changing  $\alpha$ , the strength of the steering signal, affects disagreement between steered and plain CLIP predictions. Using the profiled heads selected in our previous experiment, we perform an  $\alpha$  sweep and measure both overall classification accuracy and prediction disagreement relative to plain CLIP as seen in Figure 3.

As expected, disagreements steadily increase as  $\alpha$  becomes increasingly aggressive, suggesting that our steering mechanism is working as intended. Interestingly, this differs from the trend reported in

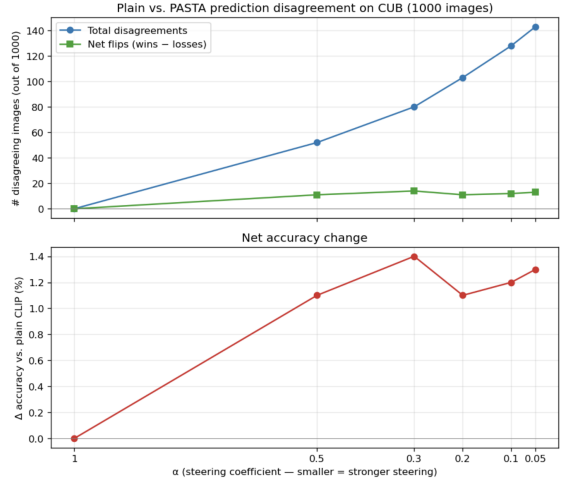


Figure 3: Effect of steering strength on classification accuracy and prediction disagreement relative to plain CLIP

| Method                        | Acc. (%) | $\Delta$ |
|-------------------------------|----------|----------|
| Plain CLIP (ViT-B/16)         | 47.68    | –        |
| Steered CLIP (random patches) | 46.87    | −0.81    |
| Steered CLIP (VLM-guided)     | 47.88    | +0.20    |

Table 2: Random-patch ablation on CUB-200-2011 (N=2500).

the original PASTA paper for LLMs, where task-following performance was largely invariant to the choice of  $\alpha$ , and highlights potential differences in steering behavior between text and visual domains.

Notably, the graph highlights that net flips, defined as wins minus losses, as well as total classification accuracy begin to plateau around  $\alpha \approx 0.3$ . This trend suggests that beyond this threshold, additional disagreements may reflect a steering strength that does not add meaningful discriminative signal between classes, and instead introduces noise.

Finally, we seek further clarification on the effectiveness of our VLM-guidance pipeline. To test whether the selected patches provide meaningful semantic guidance beyond a generic attention modification, we compare our VLM-guided patch selection mechanism against a random-patch selection baseline. For each image, the random baseline selects the same number of patch tokens as our grounding pipeline, but samples them uniformly from the 14×14 ViT patch grid. We then apply steering to the same profiled heads selected in our first experiment, while opting for an  $\alpha$  value of 0.05 to emphasize steering signal. Results are recorded in Table 2.

We see a 0.81 percentage-point decrease in per-

formance relative to standard CLIP with our arbitrary patch-selection method, resulting in a 1.01 percentage-point gain for VLM-guided steering over the random-patch baseline. Disagreement between VLM-guided CLIP and arbitrary patch selection is recorded at 13.36 percent. While random patch steering exhibits better than expected performance, likely due to a relatively small 14x14 patch grid, a 1 percentage point decrease in accuracy in comparison to our guided steering strategy still suggests that modest improvement is made through our pipeline.

## 6 Adapting CLIP To VQA

Unlike object classification, visual question answering often involves scenes containing multiple salient objects. Steering attention toward particular regions may therefore be especially useful, as it can help a vision-language model prioritize evidence most relevant to the proposed question.

CLIP does not have a native ability to support VQA tasks, as it produces image-text similarity scores rather than free-form answers. Therefore, To evaluate CLIP on a subset of the VQAv2 benchmark, we follow the methodology introduced in prior works (Song et al., 2022), reformulating yes-or-no questions as a binary image-text selection problem.

### 6.1 Dataset

We evaluate CLIP on a subset of VQAv2, a large-scale visual question answering benchmark containing real-world images paired with human-written questions and answers. VQAv2 requires models to combine visual recognition, language understanding, and common sense reasoning to answer questions about image content (Goyal et al., 2017). We focus exclusively on yes/no questions in the dataset.

### 6.2 VQA Guidance Pipeline

For each yes/no image-question pair, we prompt a VLM, Gemma 4-4b, to generate two candidate text prompts - one corresponding to a “yes” answer and one corresponding to a “no” answer. We then use the same VLM to identify up to three visual attributes or regions that would be useful for answering the question, without giving the VLM access to the corresponding image. The generated attributes are paired with the test image and passed into the text-to-region grounding model. The resulting grounded regions are converted into patch-level

steering targets using the same patch-selection and attention-steering procedure described in the previous sections. To guide the generation process, we provide two in-context learning examples of an image and question pair from prior experiments. Each example is paired with a positive attribute list that improved prediction accuracy and a negative attribute list containing broader, less question-specific visual features. Additionally, we provide 4 text-only question and attribute list examples that increased decision making accuracy in previous experiments.

We begin with a head-profiling phase, using a disjoint subset of 500 yes/no question examples from our test set. We follow the same profiling methodology described in the classification section. Each attention head in the selected transformer block is steered individually, and the top-K heads that produce the strongest improvement on the profiling subset are selected for inference.

| Method                     | Acc. (%) | $\Delta$ |
|----------------------------|----------|----------|
| Plain CLIP (CLIP ViT-B/16) | 62.10    | –        |
| Steered CLIP               | 61.10    | –1.00    |

Table 3: Zero-shot yes/no accuracy on VQAv2 val (N=1000).  $\Delta$  is relative to Plain CLIP. Steered Clip uses the top-20 profiled heads with  $\alpha = 0.1$ .

### 6.3 Results and Discussion

Overall, we observe a similar pattern to our classification experiments. Our steering method produces moderate disagreement with plain CLIP (10 percent of the test set), but the gains and losses largely cancel out, resulting in a slightly decreased overall prediction accuracy. While much of our previous classification analysis carries over to VQA, we note that some questions in the test set may not have an obvious localized visual signal to steer toward. For example, questions such as “Is the photo black and white?” or “Was this taken at night?” depend on global image properties rather than specific object regions. In these cases, forcing attention toward a small set of grounded patches may be less useful, since the relevant evidence is distributed across the entire image.

To test this hypothesis further, we take a subset of our dataset and allow Gemma 4-4b to classify each question as either localized or globalized. Localized questions require an answer that is dependent on a specific object, attribute, or relationship,

while globalized questions have answers that depend on the whole image or scene. We measure the accuracy discrepancy between the two question types:

| Question Type      | Plain CLIP | Steered CLIP | $\Delta$ |
|--------------------|------------|--------------|----------|
| Localized (N=344)  | 60.76      | 59.01        | -1.74    |
| Globalized (N=156) | 67.31      | 65.38        | -1.92    |

Table 4: Yes/no accuracy on VQAv2 validation examples split by question type. Steered CLIP uses the top-10 profiled heads with  $\alpha = 0.1$ .  $\Delta$  is relative to Plain CLIP.

Although the results align with our hypothesis in terms of delta, random noise can not be ruled out given the small performance difference between the two groups.

## 7 Limitations

We identify several potential weaknesses within our VLM-guidance pipeline and inference time steering approach. First, the  $14 \times 14$  patch grid limits how precisely our method can localize visual evidence, making it difficult to isolate small object parts or subtle attributes. Similarly, our grounding step relies on bounding boxes to select relevant image regions. Bounding boxes often include surrounding background or irrelevant neighboring objects along with the target region. A segmentation-based approach could provide more precise region selection by isolating the object or attribute more directly, but would also introduce additional computational costs.

We also recognize that CLIP’s pretrained image-text alignment may not always match human or VLM intuition about which regions are most important in accomplishing a particular task. While our pipeline assumes that steering attention toward semantically meaningful or discriminative object parts should improve recognition, CLIP may instead rely on broader visual context seen in its large pre-training corpus, including background cues that are correlated with certain classes or objects.

Finally, our pipeline is highly sensitive to the quality of the VLM-generated attributes. The generated regions may be plausible but not truly discriminative, especially when classes share similar features, or the VLM does not have relevant context regarding a classes visually similar neighbors. This can cause the steering mechanism to amplify reasonable-looking but unhelpful visual evidence, limiting its overall effectiveness, and possibly hurt-

ing CLIP’s classification ability in the process. Introducing a separate, automated judging or human feedback mechanism, along with an iterative correction process could better ensure that the selected attributes are both visually grounded and truly discriminative for the target class.

## 8 Future Work

Overall, we believe future work could extend our methodology in several directions. First, the head-profiling phase in our current experiments is performed on a relatively small subset of the target dataset. While this provides an efficient way to identify attention heads that respond positively to the steering signal, it may also lead to overfitting. Future work could explore a larger and more diverse profiling stage that incorporates examples from multiple datasets or task domains. This would more closely resemble the original PASTA methodology and may help identify attention heads that generalize better across different types of fine-grained recognition and VQA tasks.

Additionally, our method uses a fixed steering strength  $\alpha$  across all examples. However, the optimal amount of steering likely varies depending on the image, class, or question. Some examples may benefit from strong localized attention when the relevant evidence is concentrated in a small object region or distinctive pattern, while others may require weaker steering in order to preserve global context. Future work could therefore treat  $\alpha$  as a dynamic variable, either learned through lightweight training methods or predicted from the input image and patch selection phase.

## 9 Conclusion

Although our methodology produced limited improvements on both target tasks, we hoped to provide useful insight into the challenges of adapting inference-time attention steering to CLIP’s vision transformer backbone. Across our experiments, we found that increasing the steering strength,  $\alpha$ , increasingly produced disagreements with plain CLIP, although these changed predictions did not consistently translate into higher accuracy. We also introduce a VLM-guided steering pipeline that can serve as a foundation for future improvement. By generating visual attributes, grounding them into image regions, converting those regions into patch-level targets, and applying attention steering at selected heads, our method provides a modu-

lar framework for injecting external guidance into CLIP without additional fine-tuning. Finally, While our results show that directly transferring PASTA-style attention steering to the visual domain is challenging, they also identify key factors that future methods can improve on, such as patch selection, head profiling, and adaptive steering strength.

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